$V \xrightarrow{T} W$ V, W both have inner product. If $T \longleftrightarrow A$

$V \xleftarrow{T^*} W$ $\langle v, T^*(w) \rangle_V = \langle T(v), w \rangle_W$ T^* is characterised by this.

$(Av)^t \cdot w = v^t \cdot (A^t w)$

MA1522 Linear Algebra for Computing

Lesson 18: Markov Chains and Singular Value Decomposition

Suppose A is symmetric. Write $\det(x]-A) = (x-\lambda_1)^{k_1} \cdots (x-\lambda_s)^{k_s}$.

Fact 1: This splits over $\mathbb{R}$

Spectral decomposition: $\mathbb{R}^n = E_{\lambda_1} \oplus E_{\lambda_2} \oplus \cdots \oplus E_{\lambda_s}$

Fact 2: it is diagonalizable

In fact, this is an orthogonal direct sum. Fact 3: it is orthogonal.

Then take orthonormal basis S_i for each E_{λ_i}. Then $S = \bigcup_i S_i = \{v_1, \ldots, v_n\}$ is an orthonormal basis. $P = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix}$.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson18` (e.g. `A1234567Z_lesson18.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 18**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

Pollev



Section 6.4 Powers of Diagonalizable Matrices

Theorem

If

$$P^{-1}AP = D \quad \text{or} \quad A = P \begin{pmatrix} \mu_1 & 0 & \cdots & 0 \\ 0 & \mu_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mu_n \end{pmatrix} P^{-1},$$

then for every positive integer k,

$$A^k = P \begin{pmatrix} \mu_1^k & 0 & \cdots & 0 \\ 0 & \mu_2^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mu_n^k \end{pmatrix} P^{-1}.$$

If A is invertible, this holds for every integer $k \in \mathbb{Z}$.

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD^2P^{-1}$$

$$\vdots$$

$$A^k = \underbrace{(PDP^{-1})\cdots(PDP^{-1})}_{k \text{ times}} = PD^kP^{-1}$$

Section 6.4 Markov Chains

$\begin{pmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \end{pmatrix}$ $\begin{pmatrix} \frac{1}{3} \\ 0 \\ \frac{2}{3} \end{pmatrix}$

Definition

1. A vector $\mathbf{v} = (v_i) \in \mathbb{R}^n$ with nonnegative coordinates that add up to 1 is a **probability vector**.
2. A **stochastic matrix** is a square matrix whose columns are probability vectors.
3. A **Markov chain** is a sequence of probability vectors $\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_k, \ldots$ together with a stochastic matrix A such that

$$\mathbf{x}_1 = A\mathbf{x}_0, \quad \mathbf{x}_2 = A\mathbf{x}_1, \quad \ldots, \quad \mathbf{x}_k = A\mathbf{x}_{k-1}, \quad \ldots$$

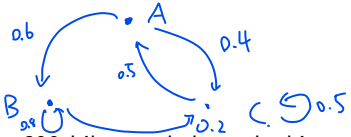
   The vector $\mathbf{x}_k$ is called a **state vector**, and $\mathbf{x}_k = A^k\mathbf{x}_0$.

4. An **equilibrium vector** is a probability vector $\mathbf{v}$ such that

$$A\mathbf{v} = \mathbf{v}.$$

   Namely, an eigenvector of A of eigenvalue 1.

Question 1



A bike-sharing scheme has 600 bikes and three docking stations A, B, C. Each day, every bike is ridden once and returned to a station:

- a bike taken from A is returned to B with probability 0.6 and to C with probability 0.4;
- a bike taken from B is returned to B with probability 0.8 and to C with probability 0.2;
- a bike taken from C is returned to A with probability 0.5 and to C with probability 0.5.

Today there are 300 bikes at A, 300 at B and none at C. Let $\mathbf{x}_k = (a_k, b_k, c_k)^T$ be the expected numbers of bikes at A, B, C after k days, so $\mathbf{x}_0 = (300, 300, 0)^T$.

1. Find the stochastic matrix A such that $\mathbf{x}_{k+1} = A\mathbf{x}_k$, and find $\mathbf{x}_1$ and $\mathbf{x}_2$.
2. Given the diagonalization of A on page 8, find the number of bikes at each station in the long run.

Question 1 — solution: part 1

- $\begin{pmatrix} a_k \\ b_k \\ c_k \end{pmatrix}$ We want to find a stochastic matrix A s.t.

$$A \begin{pmatrix} a_k \\ b_k \\ c_k \end{pmatrix} = \begin{pmatrix} a_{k+1} \\ b_{k+1} \\ c_{k+1} \end{pmatrix} = \begin{pmatrix} 0 & 0 & + 0.5c_k \\ 0.6a_k + 0.8b_k & & 0 \\ 0.4a_k + 0.2b_k & + 0.5c_k & \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 & 0.5 \\ 0.6 & 0.8 & 0 \\ 0.4 & 0.2 & 0.5 \end{pmatrix} \begin{pmatrix} a_k \\ b_k \\ c_k \end{pmatrix} \quad //$$

- $X_0 = \begin{pmatrix} 300 \\ 300 \\ 0 \end{pmatrix}$ not a probability vector.

$$X_1 = AX_0 = \begin{pmatrix} 0 \\ 420 \\ 180 \end{pmatrix}$$

$$X_2 = AX_1 = A^2X_0 = \begin{pmatrix} 90 \\ 336 \\ 174 \end{pmatrix}$$

Question 1 — part 2

2. The matrix A of Question 1 is diagonalized by

$$A = PDP^{-1}, \quad P = \begin{pmatrix} 1 & 1 & 5 \\ 3 & -2 & -3 \\ 2 & 1 & -2 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{5} \end{pmatrix}, \quad P^{-1} = \frac{1}{42}\begin{pmatrix} 7 & 7 & 7 \\ 0 & -12 & 18 \\ 7 & 1 & -5 \end{pmatrix}.$$

$A x_{k-1} = x_k$

$A^2 x_{k-2} = \cdots = A^k x_0$

Question: What is a formula for x_k?

Ans: $A^k x_0 \overset{\text{Thm}}{=\!=} (P D^k P^{-1}) x_0 = \begin{pmatrix} 1 & 1 & 5 \\ 3 & -2 & -3 \\ 2 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1^k & & \\ & (\frac{1}{2})^k & \\ & & (-\frac{1}{5})^k \end{pmatrix} P^{-1} \begin{pmatrix} 300 \\ 300 \\ 0 \end{pmatrix} = x_k.$

($A^k = P D^k P^{-1}$)

In the long run: $\lim_{k\to\infty} x_k$

$\left(\lim_{k\to\infty} (\frac{1}{2})^k = 0 \quad \lim_{k\to\infty} (-\frac{1}{5})^k = 0 \right)$

$= \begin{pmatrix} 1 & 1 & 5 \\ 3 & -2 & -3 \\ 2 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \frac{1}{42} \begin{pmatrix} 7 & 7 & 7 \\ 0 & -12 & 18 \\ 7 & 1 & -5 \end{pmatrix} \cdot \begin{pmatrix} 300 \\ 300 \\ 0 \end{pmatrix} = \begin{pmatrix} 100 \\ 200 \\ 200 \end{pmatrix}$

$= \begin{pmatrix} 1 & 0 & 0 \\ 3 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}$

entries sum up to 600.

$\in \mathbb{R}^3$.

$\operatorname{Col}\left(\begin{pmatrix} 1 & 0 & 0 \\ 3 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}\right) = \operatorname{span}\left\{\begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}\right\}$.

Recall.

$\operatorname{Col}(A) = \{ Av \mid v \in \mathbb{R}^n \}$.

Section 6.4 Regular Stochastic Matrices

Definition

A stochastic matrix A is **regular** if A^k has all entries positive for some positive integer k.

Theorem

If A is a regular stochastic matrix, then A has a unique equilibrium vector $\mathbf{v}$, and for every probability vector $\mathbf{x}_0$, the Markov chain $\mathbf{x}_k = A^k\mathbf{x}_0$ converges to $\mathbf{v}$.

If A is regular then

a) The geometric mult of 1 is 1.

b) All the other eigenvalues λ_i has $|\lambda_i| < 1$.

Algorithm to compute an equilibrium vector.

1. Find a nontrivial solution $\mathbf{u} = (u_i)$ of $(I - A)\mathbf{x} = \mathbf{0}$.
2. Then $\mathbf{v} = \dfrac{1}{u_1 + u_2 + \cdots + u_n}\mathbf{u}$ is an equilibrium vector.

Question 2

Let A be the matrix in Question 1,

$$A = \begin{pmatrix} 0 & 0 & 0.5 \\ 0.6 & 0.8 & 0 \\ 0.4 & 0.2 & 0.5 \end{pmatrix}.$$

A non-regular stochastic matrix:

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

$$A^k = \begin{cases} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & \text{if } k \text{ is odd} \\ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & \text{if } k \text{ is even.} \end{cases}$$

1. Show that A is regular.

$$A^2 = \begin{pmatrix} 0.2 & 0.1 & 0.25 \\ 0.48 & 0.64 & 0.3 \\ 0.32 & 0.26 & 0.45 \end{pmatrix} \Rightarrow A \text{ is regular.}$$

2. Find the equilibrium vector of A.

The unique eigenvector of A with eigenvalue 1. s.t. $x+y+z=1$

$\begin{pmatrix} x \\ y \\ z \end{pmatrix}$

From previous question, $E_1 = \text{span}\left\{\begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}\right\}$.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{6}\begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}.$$

Question 2 — solution

Challenge

The **Fibonacci numbers** are defined by

$$F_0 = 0, \qquad F_1 = 1, \qquad F_{n+2} = F_{n+1} + F_n \quad (n \geq 0),$$

so the sequence is $0, 1, 1, 2, 3, 5, 8, 13, \ldots$. Let $A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$.

1. Show that $A \begin{pmatrix} F_n \\ F_{n+1} \end{pmatrix} = \begin{pmatrix} F_{n+1} \\ F_{n+2} \end{pmatrix}$ for every $n \geq 0$.
2. Diagonalize A, and show that for every $n \geq 0$,

$$F_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right].$$

Hint: by part 1, $A^n \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} F_n \\ F_{n+1} \end{pmatrix}$.

Section 6.5 Singular Value Decomposition

Definition

Let A be an $m \times n$ matrix. A **singular value decomposition** of A is an expression

$$A = U\Sigma V^T,$$

where

- U and V are $m \times m$ and $n \times n$ orthogonal matrices;
- Σ is an $m \times n$ "diagonal matrix":

$$\Sigma = \begin{pmatrix} D & \mathbf{0}_{r\times(n-r)} \\ \mathbf{0}_{(m-r)\times r} & \mathbf{0}_{(m-r)\times(n-r)} \end{pmatrix}, \qquad D = \operatorname{diag}(\sigma_1, \ldots, \sigma_r), \quad \sigma_1 \geq \cdots \geq \sigma_r > 0.$$

Theorem

Every $m \times n$ matrix A has a singular value decomposition. Here $r = \operatorname{rank}(A)$, and $\sigma_1 \geq \cdots \geq \sigma_n$ are square roots of the eigenvalues of $A^T A$ (the **singular values** of A).

Section 6.5 Finding a Singular Value Decomposition

Suppose that $A = U\Sigma V^T$ is a singular value decomposition. (A: $m \times n$; V^T: $n \times n$.)

- $A^TA = (U\Sigma V^T)^T(U\Sigma V^T) = V\Sigma^T(U^TU)\Sigma V^T = V\Sigma^T\Sigma V^T = V\begin{pmatrix}\sigma_1^2 & & & & \\ & \sigma_2^2 & & & \\ & & \ddots & & \\ & & & \sigma_r^2 & \\ & & & & 0 \end{pmatrix}V^T$ ($U^TU = I$).

  So $V = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix}$ orthogonally diagonalize A^TA.

  $\Sigma = \begin{pmatrix} \sigma_1 & & & \\ & \ddots & & \\ & & \sigma_r & \\ & & & 0 \end{pmatrix}\Big\}m$ where $\sigma_1^2 \geq \cdots \geq \sigma_r^2 > 0 = \cdots = 0$ are eigenvalues of A^TA.

- $A = U\Sigma V^T \longrightarrow AV = U\Sigma = \begin{pmatrix} \sigma_1u_1 & \cdots & \sigma_ru_r & 0 & \cdots & 0 \end{pmatrix}$

  $U = (u_1, \ldots, u_r, u_{r+1}, \ldots, u_m)$ So $u_i = \frac{1}{\sigma_i}Av_i$ for $1 \leq i \leq r$

- Extend $\{u_1, \ldots, u_r\}$ to an orthonormal basis of $\mathbb{R}^m$. $\{u_1, \ldots, u_r, u_{r+1}, \ldots, u_m\} \rightsquigarrow U$.

Section 6.5 Algorithm for Singular Value Decomposition

Let A be an $m \times n$ matrix of rank r.

1. Find the eigenvalues $\mu_1 \geq \cdots \geq \mu_r > 0 = \mu_{r+1} = \cdots = \mu_n$ of $A^T A$; let $\sigma_i = \sqrt{\mu_i}$ and form Σ.
2. Find an orthonormal basis of each eigenspace of $A^T A$; let $\mathbf{v}_i$ be the unit eigenvector for μ_i, and $V = \begin{pmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{pmatrix}$.
3. Let $\mathbf{u}_i = \frac{1}{\sigma_i} A\mathbf{v}_i$ for $i = 1, \ldots, r$. Extend to an orthonormal basis $\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$ of $\mathbb{R}^m$ by solving $\begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_r \end{pmatrix}^T \mathbf{x} = \mathbf{0}$, and let $U = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_m \end{pmatrix}$.

Exercise.

1. Show that the eigenvalues of the symmetric matrix $A^T A$ are nonnegative.
2. Show that in step 3, $\{A\mathbf{v}_1, \ldots, A\mathbf{v}_r\}$ is orthogonal.

Question 3

Find a singular value decomposition of

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \\ -2 & 0 \end{pmatrix}.$$

That is, find orthogonal matrices U (of order 3) and V (of order 2), and the 3×2 matrix Σ, such that $A = U\Sigma V^T$.

Question 3 — solution: Σ and V

$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \\ -2 & 0 \end{pmatrix}$.

Step 1 Orthogonally diagonalize A^TA

$A^TA = \begin{pmatrix} 2 & 0 & -2 \\ 1 & 2 & 0 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 0 & 2 \\ -2 & 0 \end{pmatrix} = \begin{pmatrix} 8 & 2 \\ 2 & 5 \end{pmatrix}$

$\det(xI_2 - A^TA) = \det\begin{pmatrix} x-8 & -2 \\ -2 & x-5 \end{pmatrix} = (x-8)(x-5) - 4$
$= x^2 - 13x + 36.$
$= (x-4)(x-9).$

- Find orthonormal basis for each eigenspace.

$E_4 = \text{Null}(4I - A^TA)$
$= \text{Null}\begin{pmatrix} -4 & -2 \\ -2 & -1 \end{pmatrix} = \text{span}\left\{\begin{pmatrix} 1 \\ -2 \end{pmatrix}\right\}$.

$\Rightarrow v_2 = \frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ -2 \end{pmatrix}$

$E_9 = \text{Null}\begin{pmatrix} 1 & -2 \\ -2 & 4 \end{pmatrix} = \text{span}\left\{\begin{pmatrix} 2 \\ 1 \end{pmatrix}\right\}$.

$v_1 = \frac{1}{\sqrt{5}}\begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

$\Longrightarrow V = \begin{pmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{-2}{\sqrt{5}} \end{pmatrix}$ ($\leftrightarrow 9$, $\leftrightarrow 4$; $= v_1$, $= v_2$)

$\Sigma = \begin{pmatrix} 3 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix}$ ($3 = \sqrt{9} = \sqrt{\lambda_1}$, $\neq \sqrt{\lambda_2}$).

$r = 2$.

Question 3 — solution: U

Step 2: $u_i = \frac{1}{\sigma_i} A v_i$ for $1 \le i \le r$. $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \\ -2 & 0 \end{pmatrix}$

$u_1 = \frac{1}{3} A \cdot \begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{pmatrix} = \frac{1}{3\sqrt{5}} \begin{pmatrix} 2 & 1 \\ 0 & 2 \\ -2 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \frac{1}{3\sqrt{5}} \begin{pmatrix} 5 \\ 2 \\ -4 \end{pmatrix}$.

$u_2 = \frac{1}{2} A \cdot \begin{pmatrix} \frac{1}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} \end{pmatrix} = \frac{1}{2\sqrt{5}} \begin{pmatrix} 2 & 1 \\ 0 & 2 \\ -2 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \end{pmatrix}$

$= \frac{1}{2\sqrt{5}} \begin{pmatrix} 0 \\ -4 \\ -2 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 0 \\ -2 \\ -1 \end{pmatrix}$

Step 3 Extend $\{u_1, u_2\}$ to an orthonormal basis of $\mathbb{R}^3$.

Apply Gram-Schmidt to $\{u_1, u_2, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\}$.

$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \rightsquigarrow \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - \frac{u_1 \cdot e_1}{u_1 \cdot u_1} u_1 - \frac{u_2 \cdot e_1}{u_2 \cdot u_2} \cdot u_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - \frac{5}{3\sqrt{5}} \cdot \frac{1}{3\sqrt{5}} \begin{pmatrix} 5 \\ 2 \\ -4 \end{pmatrix} - 0$

$= \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - \frac{1}{9} \begin{pmatrix} 5 \\ 2 \\ -4 \end{pmatrix} = \frac{1}{9} \begin{pmatrix} 4 \\ -2 \\ 4 \end{pmatrix} = u_3'$

$\|u_3'\|^2 = \frac{1}{81}(16+4+16) = \frac{36}{81} = \frac{4}{9} = \left(\frac{2}{3}\right)^2$

$u_3 = \frac{u_3'}{\|u_3'\|} = \frac{3}{2} \times \frac{1}{9} \begin{pmatrix} 4 \\ -2 \\ 4 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$

$U = \begin{pmatrix} | & | & | \\ u_1 & u_2 & u_3 \\ | & | & | \end{pmatrix}$

Homework

1. Search for applications of the singular value decomposition.

2. Find a singular value decomposition of each of the following matrices.

$$H_1 = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}, \qquad H_2 = \begin{pmatrix} 2 & 2 \\ 1 & -1 \end{pmatrix}, \qquad H_3 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 0 \end{pmatrix},$$

$$H_4 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \qquad H_5 = \begin{pmatrix} 2 & 1 \\ 2 & 1 \\ 2 & 1 \end{pmatrix}.$$

Next lesson

Next lesson (**Fri 30 Oct**, Section 7.1), I will focus on

1. linear transformations, and

2. the standard matrix of a linear transformation.

Please watch the video clips beforehand.